

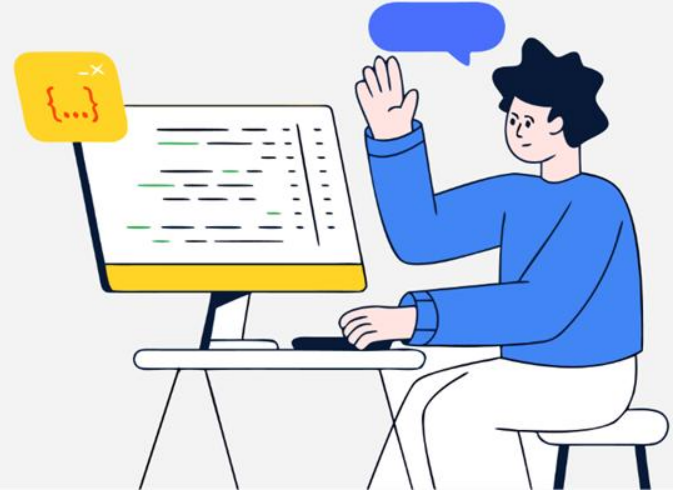
# Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.
- In case of queries, kindly reach out to us at [solutionchallengesupport@hack2skill.com](mailto:solutionchallengesupport@hack2skill.com)

**Note:** Prototype must include cloud deployment and use at least one Google AI model or service (e.g., Gemini) as part of the solution.



# Solution Challenge



## Team Details

**Team name:** double trouble 2x

**Team members:**

- 1) Aadi shah
- 2) nitya singh

**Problem Statement:** creators, journalists, marketplaces, and moderators need a fast way to detect duplicate media and verify ownership before content is reused, sold, or published.

## Brief about SecureMedia AI

SecureMedia AI is a lightweight authenticity workflow for image uploads. The product combines perceptual hashing, optional AI similarity, and blockchain ownership status in one review step.

The MVP now runs as a live Google Cloud deployment with a React upload interface, a Flask upload API, duplicate detection, ownership verification fields, and a minimal smart contract path for EVM testnets.

For Google Solution Challenge alignment, the production path now uses Cloud Run hosting and is ready for Vertex AI multimodal similarity as the Google-native scoring layer.

### Trust

Hash + AI  
+  
ownership

### Live

Cloud Run  
deployment

### Next Step

Vertex AI scoring  
Managed secrets

## Opportunities and Differentiation

### Different from similar ideas

Combines duplicate detection and ownership tracking in one workflow.

Modular services keep the stack lightweight for pilots and hackathons.

Still degrades gracefully when AI or blockchain modules are unavailable.

### How it solves the problem

Flags suspicious duplicate uploads before publication or resale.

Adds ownership context to moderation and review decisions.

Reduces ambiguity for creator-support, newsroom, and marketplace teams.

### USP

One API response with similarity, duplicate state, owner, and verification.

Challenge-ready path to Google AI and Google Cloud.

Simple enough to ship, strong enough to extend.

## List of Features Offered by the Solution

### Image upload

Drag-and-drop or picker-based image submission

### Live preview

Visual confirmation before analysis begins

### Similarity score

Clear percentage output for fast review

### Duplicate status

Original vs duplicate decision signal

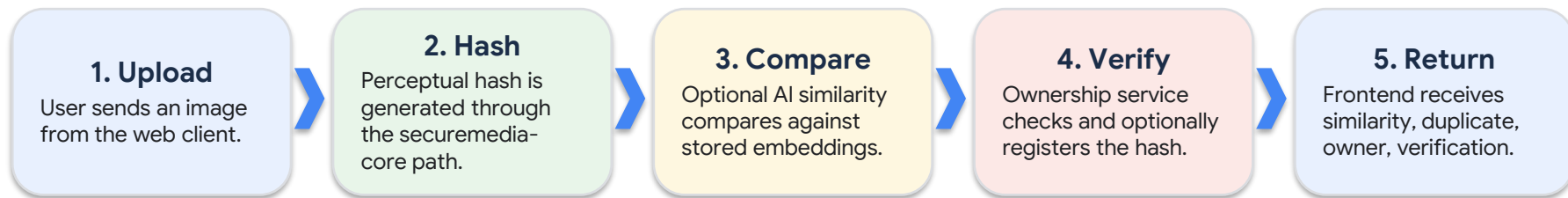
### Ownership status

Owner string and blockchain verification field

### Modular backend

Hashing, AI, and blockchain modules work independently

## Process Flow Diagram



Primary users: creators, moderators, marketplaces, journalists, and authenticity review teams.

## Wireframes / Mock Diagrams of the Proposed Solution

The wireframe shows a web interface for 'SECUREMEDIA AI' with the title 'Upload media'. Below the title is a subtitle: 'Preview the image, send it to the Flask API, and review similarity, duplicate status, owner, and blockchain verification in one place.' A progress indicator shows 'Awaiting analysis'.

The interface is divided into two main sections:

- File upload:** Contains a large light blue box with a '+' icon and the text 'Choose image'. Below this, it says 'PNG, JPG, WEBP, or GIF files work best for secure-media checks.' At the bottom of this section are two buttons: 'Clear selection' and 'Analyze image'.
- Result:** Contains a progress indicator 'Not verified' and four status cards:
  - Similarity:** Shows '--'.
  - Duplicate status:** Shows 'Awaiting upload'.
  - Owner:** Shows 'Unverified'.
  - Blockchain:** Shows 'Not verified' in red text.

### Upload panel

Single image input, preview area, and analyze trigger.

### Decision cards

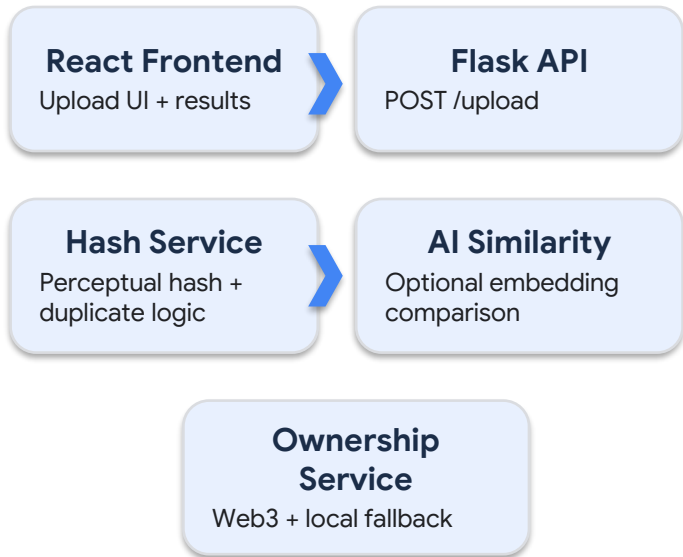
Similarity, duplicate status, owner, and blockchain verification stay visible together.

### Review posture

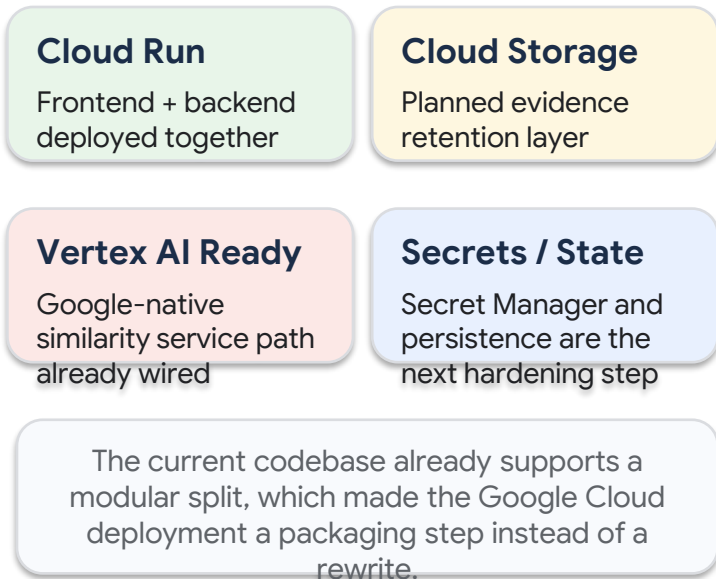
Optimized for quick operator decisions, not consumer browsing.

## Architecture Diagram of the Proposed Solution

### Current MVP



### Live Google Cloud Deployment





## Technologies to be Used in the Solution

### Current Prototype Stack

React 19 + Vite

Tailwind CSS

Flask API

Pillow + ImageHash

Optional Hugging  
Face CLIP

Solidity + Web3

Sepolia / Polygon  
testnets

### Challenge Deployment Stack

Cloud Run (live)

Cloud Storage

Secret Manager

Vertex AI  
multimodal  
embeddings

Cloud Build /  
Artifact Registry

Monitoring +  
Logging

## Estimated Implementation Cost

### Estimated monthly cost for the live pilot on Google Cloud

| Component            | Assumption                                          | Monthly cost |
|----------------------|-----------------------------------------------------|--------------|
| Cloud Run            | Live frontend + Flask API service in us-central1    | \$10-\$25    |
| Cloud Storage        | Uploaded image retention and evidence objects       | \$2-\$5      |
| Vertex AI            | Similarity calls when Google scoring is switched on | \$20-\$40    |
| Observability        | Basic logs, traces, and monitoring                  | \$0-\$10     |
| Testnet transactions | Registration on Sepolia / Amoy                      | Negligible   |

**Estimated pilot total: about \$32-\$80 per month, with hackathon credits likely reducing early cost.**

# Snapshots of the MVP

## Live Cloud Run deployment

SECUREMEDIA AI

### Upload media

Preview the image, send it to the Flask API, and review similarity, duplicate status, owner, and blockchain verification in one place.

Awaiting analysis

**File upload**

Choose one image to run similarity, duplicate, and ownership checks.

+

Choose image

PNG, JPG, WEBP, or GIF files work best for secure-media checks.

No image selected. Clear selection Analyze image

**Result**

Similarity, duplicate state, owner, and blockchain verification.

Not verified

Similarity

--

Duplicate status

Awaiting upload

Owner

Unverified

Blockchain

Not verified

## Local operator flow with file selected

SECUREMEDIA AI

### Upload media

Preview the image, send it to the Flask API, and review similarity, duplicate status, owner, and blockchain verification in one place.

Awaiting analysis

**File upload**

Choose one image to run similarity, duplicate, and ownership checks.

sample-upload.png

0.1 KB

Clear selection Analyze image

**Result**

Similarity, duplicate state, owner, and blockchain verification.

Not verified

Similarity

--

Duplicate status

Awaiting upload

Owner

Unverified

Blockchain

Not verified

## Production smoke-test response

```
{ "similarity": 0.0, "duplicate": false, "owner": "Unverified", "blockchain_verified": false }
```

## Additional Details / Future Development

### Immediate engineering follow-up

Turn on Vertex AI scoring in production with a managed embedding store.

Replace placeholder chain config with a real contract deployment and funded wallet.

Persist uploads, ownership history, and review trails beyond local JSON fallback.

### Challenge-ready roadmap

Add creator accounts, evidence history, and case management.

Introduce Secret Manager and Cloud-native persistence for production operations.

Extend from image uploads to video frames, watermarking, and newsroom workflows.

## Project Links

### GitHub Public Repository

<https://github.com/Aadishah17/secure-media>

### Demo Video Link (3 Minutes)

To be recorded before submission

### Live MVP Link

<https://securemedia-ai-97902534410.us-central1.run.app>

### Health Check

<https://securemedia-ai-97902534410.us-central1.run.app/api/health>

# Impact and SDG Alignment

## Why it matters

Protects original creators from silent reuse and counterfeit claims.

Gives moderators stronger evidence before approving or rejecting media.

Builds a path toward safer, more trustworthy visual publishing systems.

## SDG 9

Industry, Innovation and Infrastructure

## SDG 16

Peace, Justice and Strong Institutions

## SecureMedia AI

Protect originality. Prove ownership.

[GitHub: github.com/Aadishah17/secure-media](https://github.com/Aadishah17/secure-media)

[Live app: securemedia-ai-97902534410.us-central1.run.app](https://securemedia-ai-97902534410.us-central1.run.app)

